Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Python software engineer with 7+ years building production systems for multifamily property management software - Entrata: 100K+ monthly active users, 1M+ property records, HUD/LIHTC affordable housing compliance. Currently at Intuit integrating Anthropic Claude into financial applications using Claude Code as the primary development interface. Built LeaseI, an 8-agent LangGraph platform for lease compliance with human-in-the-loop escalation (93.6% precision, 1,240 test cases), and AgentProbe, an open-source LLM evaluation toolkit covering Claude API and four other providers. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Portfolio: venkatalakshmiparimala.github.io/Portfolio

CORE COMPETENCIES

Claude Code | Claude API Pipelines | MCP Integrations | Agentic Workflows | Human-in-the-Loop (HITL) | LangGraph / LangChain | LLM Evaluation Frameworks | Prompt Engineering | Multifamily Property Management | PostgreSQL | AWS | REST APIs | FastAPI / Python | RAG Pipelines | GitHub CI/CD

EXPERIENCE

Intuit

Jun 2026 - Present

Software Engineer · Virginia, USA

- Integrate Anthropic Claude into Python/FastAPI services using **Claude Code as the primary development interface** - financial data summaries, natural-language queries, and reporting workflows.
- Design prompt chains and structured-output schemas for Claude so financial summaries stay consistent and validate cleanly in downstream services.
- Build and maintain **MCP-compatible tool integrations** for structured data access across services.
- Ground Claude responses in customer financial data via retrieval so every answer traces back to a source record.
- Build LLM evaluation and regression tests (LLM-as-judge + golden datasets) to catch hallucinations and quality drops before release.
- Add guardrails around LLM calls: input validation, sensitive-data handling, and fallbacks when model output fails checks.
- Build ML pipelines (scikit-learn, XGBoost) from data ingestion and feature engineering to prediction APIs; containerize and ship via GitHub Actions CI/CD to AWS.

Entrata - Multifamily & Affordable Housing SaaS

Oct 2020 - Jul 2024

Senior Software Engineer · 100K+ monthly active users, 1M+ property records

- Delivered 20+ features end-to-end (Python, PHP, Java, JavaScript, React, PostgreSQL) across multifamily and affordable housing property management modules - 120+ Agile sprints.
- Owned the annual **HUD income-limit import** for 6 years: transformed federal data into ~180,000 tenant eligibility records with LIHTC/HUD validation, rollback checks, and zero compliance violations across every release cycle.
- Built a **RabbitMQ-based async processing system** for rent updates across thousands of multifamily tenant records - retry logic and idempotency eliminated double-processing failures.
- Wrote Python/Bash automation that cut manual operational work by 80%; built internal debugging tools that cut triage time by 40%.
- Optimized 200+ SQL queries using execution plans and indexing - 30% average performance gain across multitenant production database.
- Led Docker microservices redesign (50% better scalability, 30% faster deployments) and AWS S3 document migration with Libsodium encryption for resident PHI - zero security incidents.
- Reviewed 300+ pull requests; mentored junior engineers.

Entrata India

Sep 2018 - Oct 2020

Junior Software Engineer · Pune, India

- Built affordable housing and rental management modules (Python, PHP, Java, PostgreSQL) - leasing, certification, and compliance workflows for multifamily operators.
- Delivered large-scale data import and reconciliation pipelines for **1.6M HUD/Tax Credit records**.
- Led AWS S3 cloud migration from legacy server storage; created 56 state-specific housing certification documents.

PROJECTS

Leasel - Multi-Agent Lease Compliance Platform

Multifamily AI · HITL

Python · LangGraph (8-agent) · FastAPI · Streamlit · TF-IDF RAG · SQLite · HUD API · OCR - github.com/venkatalakshmiparimala/Lease_complaince_rag

Built an AI-powered lease compliance and support investigation platform for affordable housing operations. An 8-agent LangGraph workflow (triage, retrieval, reasoning, validation) investigates issues and escalates to a human when confidence is low. Core design rule enforced in code: LLM extracts and reasons; deterministic or human logic makes every eligibility, compliance, and financial call. RAG pipeline: OCR ingestion, section-aware chunking, TF-IDF hybrid retrieval, answers cited to the exact source clause. LLM-as-judge CI: 93.6% precision and 87.9% F1 across 1,240 test cases; 100% citation correctness on document Q&A.

AgentProbe - Open-Source LLM Evaluation Toolkit

Claude API · Agentic

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · Anthropic Claude · OpenAI · Azure OpenAI · Groq · Ollama - github.com/venkatalakshmiparimala/AgentProbe

Built a 15-endpoint FastAPI backend routing prompts across five LLM providers - Anthropic Claude API, OpenAI, Azure OpenAI, Groq, and Ollama. Scores LLM and RAG outputs on five metrics: factual accuracy, hallucination rate, retrieval quality, completeness, and compliance using LLM-as-judge. Three LangGraph agents (Orchestrator, RAG Tracer, Auto Fix) with CI regression tests on golden datasets. Runs 100% locally at zero cost.

AI on FHIR - Clinical NLP & LLM API Integration

HITL · LLM

Python · Flask · React · TypeScript · Docker · BERT · spaCy · LLMs · FHIR REST APIs - github.com/venkatalakshmiparimala/AI_on_FHIR

Translates English/Spanish questions into FHIR API calls using LLMs, BERT, and embeddings - cut manual clinical lookup time by ~60%. HITL routing for sensitive data, role-based access for HIPAA-style data isolation, containerized with Docker.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia May 2026

GPA 3.87 · Coursework: ML systems, NLP, data pipelines, statistical modeling, AI engineering

B.Tech Computer Science & Engineering - JNTU Anantapur 2017

TECHNICAL SKILLS

AI & LLMs: Anthropic Claude (API + Claude Code), MCP integrations, LangGraph, LangChain, agentic workflows, HITL, RAG pipelines, prompt engineering, structured outputs, LLM-as-judge evaluation, embeddings, ChromaDB, Pinecone, Weaviate, Elasticsearch

Python & Backend: Python (9+ yrs), FastAPI, Flask, Django, PostgreSQL, MySQL, Oracle, Redis, MongoDB, SQLite, REST APIs, Microservices, OAuth

Cloud & DevOps: AWS (S3, EC2, RDS, Lambda, Glue, SQS, API Gateway, IAM), Railway, Docker, Kubernetes, Terraform, GitHub Actions CI/CD, Bash

Data & ML: Pandas, NumPy, scikit-learn, XGBoost, PyTorch, PySpark, ETL pipelines, Snowflake, data modeling, query optimization

Other Languages: Java, Spring Boot, TypeScript, JavaScript, Node.js, React, PHP, SQL, GraphQL

Messaging & Monitoring: RabbitMQ, Kafka, Celery, Prometheus, Grafana, OpenTelemetry